

Confidence and Calibration of Activation Oracles for Reliable Interpretation of Language Model Internals

 **Federico Torrielli**

University of Turin
federico.torrielli@unito.it

 **Peter Schneider-Kamp**

University of Southern Denmark
petersk@imada.sdu.dk

 **Lukas Galke Poech**

University of Southern Denmark
galke@imada.sdu.dk

Abstract

An activation oracle is a language model trained to read another model’s internal activations and describe them in natural language, for example to name a secret word the other model was trained to hide. Oracle answers carry no measure of confidence, which limits their use in auditing. We compare five ways of attaching a confidence score to an oracle’s answer on this secret-word task, across four oracles from two model families (Qwen and Gemma, 8B to 27B parameters), at 6,000 samples per method and oracle. The five methods rank the same way on all four oracles. Which method to use depends on one question: can the auditor list the possible answers in advance? If the auditor can, then having the oracle score each candidate answer roughly doubles accuracy and separates correct from wrong answers best of the five (AUROC 0.92 to 0.96). If the oracle must generate its answer freely and no labeled data exists, the agreement rate over twenty samples is the only confidence that is calibrated on every oracle. Once labeled data exists, a rescaled answer probability reaches the same calibration at one generation instead of twenty. Asking the oracle to state a confidence number gives no usable signal on any oracle. Code and the patched trainer are available at https://github.com/federicotorrielli/probabilistic_activation_oracles.

1 Introduction

A large language model holds information in its internal activity while it works, and some of that information never shows up in the text it writes. An activation oracle is a second model trained to look at that internal activity and put it into words (Karvonen et al., 2025) (fig. 1, top). To test it, Karvonen et al. (2025) play a guessing game. They take a model that has been fine-tuned to keep a secret word, let the oracle look at that model’s internal activity, and ask the oracle to name the

word. Sometimes the oracle is right, sometimes it is wrong. This paper asks a different question: does the oracle know when it is right? A confidence score is called well calibrated when it matches how often the answer is in fact correct. We compare five ways of producing such a score and report which ones are calibrated.

The question matters because of what oracles are being proposed for: alignment auditing (Bricken et al., 2025; Sheshadri et al., 2026), deception detection (Ravindran, 2025), and elicitation of hidden objectives (Dietz et al., 2026). Each of these ends in a yes-or-no decision made by comparing a confidence score against a threshold: an auditor flags or releases a model, a monitoring pipeline routes or drops a generation. A miscalibrated score places that threshold wrong, so the pipeline acts on confident answers that are wrong or discards usable ones. Karvonen et al. (2025) name the lack of calibration as a main limitation of their oracles, and Jakkli et al. (2026) list missing calibration training among the reasons current oracles are hard to use on safety tasks. So far, no one has measured which confidence methods actually work on an oracle.

There are standard ways to attach a confidence score to a language model’s output, but they were developed on unmodified models. An oracle is a steered model: while it answers, vectors taken from another model are added into part of its internal state. Whether the standard methods still work under that intervention is an empirical question. We answer it on the 20-word secret-word task of Karvonen et al. (2025), on four oracles spanning two model families and a $3.3\times$ size range: Qwen3-8B and Gemma-2-9B (released by Karvonen et al., 2025), Gemma-3-27B, and a Qwen3.6-27B oracle we train and release. The five methods adapt established baselines: the probability the oracle assigned to its answer, agreement across twenty sampled answers (a short-answer form of self-consistency, Wang et al., 2022), asking the oracle for a confi-



Figure 1: An activation oracle names a hidden word and attaches no confidence to it. We compare five ways of supplying one. †: readouts introduced or first evaluated in this work. Three further readouts built for the steered setting are reported in section J.

dence number, in free text or over a fixed set of confidence labels (Kadavath et al., 2022; Lin et al., 2022), and scoring the 20 candidate words directly. We also designed three readouts that use the steering machinery itself, and all three carry weaker confidence than the five on every oracle; section J reports them with the mechanism behind each result.

The five methods rank the same way on all four oracles, so none of the advice below depends on which oracle an auditor is reading. The auditor’s own situation does vary, and it splits the advice on one question: can the auditor list the possible answers in advance? When the 20 candidate words are known, letting the oracle score each of them directly is better on every measure: accuracy roughly doubles relative to generating an answer (e.g. 0.65 vs. 0.41 on Qwen3-8B, 0.53 vs. 0.23 on Qwen3.6-27B), and the resulting confidence separates correct from wrong answers best (AUROC 0.92 to 0.96). This confirms, with calibration numbers, the conjecture of Jakkli et al. (2026) that an oracle gives up more of what it knows when its output format is constrained than when it generates freely. When the answer must be generated freely and no labeled data is available, the agreement rate over twenty sampled answers (the bootstrap, at a tuned sampling temperature) is the only confidence that is calibrated on every oracle (expected calibration error, ECE, 0.04 to 0.12); every competing method is well calibrated on some oracles and far off on

others. Once labeled data exists, a correction fitted to the answer’s probability reaches the same calibration with a single generation instead of twenty. Asking the oracle for a confidence number gives no usable signal on any oracle (AUROC 0.40 to 0.53), and on Qwen3.6-27B the stated number is on average higher when the answer is wrong. This replicates, on steered oracles, a known gap on ordinary models: the uncertainty signal is present in the activations but does not reach the model’s spoken confidence (Yuan et al., 2026; Miao and Ungar, 2026). Consistent with that reading, a constrained version of the same question, which scores five fixed confidence labels with the oracle’s own output probabilities, recovers a usable signal (best-readout AUROC 0.68 to 0.82).

Contributions.

- The first study on uncertainty quantification for activation oracles: five confidence estimation methods compared head to head, plus three further readouts built for the steered setting in section J, on four oracles from two model families, against five metrics, at 6,000 samples per method-oracle pair, with bootstrap confidence intervals and a controlled target-set-size variant. Which methods work, and how well, is stable across family and scale.
- A testbed for this question, released with the paper. It holds the four oracles, twenty secret-word target models for each, the 6,000 sample

protocol, and a scorer that takes a list of answer and confidence pairs and returns accuracy, expected calibration error, Brier score, negative log-likelihood, and AUROC with bootstrap intervals. A new confidence method joins the comparison by emitting those pairs, and the numbers reported here are what it is measured against. We also release the first activation oracle for a hybrid attention architecture, Qwen3.6-27B, together with the trainer changes it required and the forty secret-word target models we retrained.

- A practical decision rule with three cases. If the candidate answers can be listed, have the oracle score them directly. If the answer must be generated and no labeled data exists, use the agreement rate over sampled answers, with the sampling temperature chosen so that mean agreement matches accuracy on a held-out slice. Once labeled data exists, a rescaled answer probability gives the same calibration at 1/20th of the generation cost.
- A replication, on steered oracles, of the finding that activations carry an uncertainty signal the model’s spoken confidence does not (Yuan et al., 2026; Miao and Ungar, 2026), together with a constrained-label readout, evaluated at full scale on all four oracles, that recovers the signal the free-text channel loses.

2 Background and Related Work

Describing activations in words. A transformer stores information about its input in the hidden states of its residual stream. Several methods take such a hidden state and return a description of it in natural language. The logit lens (nostalgebraist, 2020) applies the model’s own output layer to a hidden state taken at an intermediate layer, and returns the resulting distribution over vocabulary tokens. The logit lens needs no training, and its description is a ranked list of tokens, so the logit lens reports only which token the model is about to predict. The activation difference lens (Minder et al., 2026) subtracts the activations of a base model from the activations of a model fine-tuned from it. Minder et al. (2026) show that a description of this difference names the topic the fine-tuning taught, and that on a model fine-tuned to hide a word it names the hidden word. The activation difference lens describes a model. The methods in this paper describe one hidden state at a time.

An activation oracle is a language model trained

to accept another model’s hidden states as input and answer questions about them in natural language. Pan et al. (2024) introduce this design: they build pairs of an activation vector and a written answer about that vector, then fine-tune a decoder model on those pairs. Karvonen et al. (2025) train a decoder the same way on a wider mixture of tasks, call the result an activation oracle, and test it on inputs far from its training data. The four oracles we benchmark are of this kind. Section 3.1 states how a hidden state reaches an oracle and how the oracle answers.

Why oracle answers need a confidence. The oracle’s output is a plain-text sentence with no notion of certainty attached. Karvonen et al. (2025) observe that their model “will frequently produce an answer even when confidence is low”. Jakkli et al. (2026) test the released Qwen3 oracles on safety-relevant tasks, find that roughly half of free-form answers are too generic to falsify, and identify the absence of calibration training as one of three fixable causes. They also conjecture that constrained output formats (multiple choice, binary, scored logits) extract more signal from an oracle than free-form generation. Our forced-choice and constrained-label results test that conjecture directly. In parallel, Bauer et al. (2026) improve the oracle training recipe and, in their evaluation suite, mitigate hallucination at inference time by sampling k answers and keeping only those above an agreement threshold. That filter is the same signal as our bootstrap method; we contribute the calibration study of it, that is, whether the agreement rate can be read as a probability. An oracle reads a target model’s hidden state by adding a rescaled copy of that hidden state to its own residual stream. Activation steering methods perform the same addition on a model’s residual stream to change how the model behaves, for example to make it more cautious (Turner et al., 2024; Rimsky et al., 2024; Zou et al., 2025). The purpose is what differs: a steering method picks the added vector to obtain a behavior, an oracle adds the hidden state in order to describe it, and the target model is left untouched.

Uncertainty for language models. Modern neural networks are systematically miscalibrated, and a simple post-hoc rescale on labeled data reduces the error (Guo et al., 2017). Post-hoc temperature scaling is a one-knob correction: on a small set of examples with known answers, find the single number that makes the reported confidences match

reality best, and divide by it. We report both native calibration (no rescale) and post-hoc calibration (section 5.4). Kadavath et al. (2022) show that large models assign well-calibrated probabilities on multiple-choice questions and that a model’s probability of the token “True”, when asked to evaluate its own proposed answer, is an informative score. Lin et al. (2022) train models to verbalize calibrated confidence, but the calibration does not transfer to held-out task families without further training; without such training, verbalized confidence is systematically overconfident (Mielke et al., 2022; Xiong et al., 2024). Wang et al. (2022) introduce self-consistency: sample k reasoning traces, take the most frequent final answer, and read the agreement rate as a confidence. Our bootstrap method is the short-answer form of this idea.

Probe versus verbalized confidence. Yuan et al. (2026) and Miao and Ungar (2026) report that the uncertainty signal present in a model’s activations is largely unreachable through the model’s spoken confidence: a linear probe on hidden states predicts correctness at AUROC up to 0.95 while the verbalized confidence is not informative about correctness, and the verbalized-confidence direction is roughly orthogonal to the calibration direction in activation space. We observe the same gap on steered oracles (section 5.5) and show that a constrained label readout, which scores a fixed set of answers with the model’s own logits instead of letting it generate, recovers part of the signal. Basu et al. (2026) report a parallel gap on a clinical task.

3 Methods

3.1 Preliminaries

We distinguish between the *target model*, which is the model under inspection, and the *oracle*, which is the model that reads the target’s hidden states and answers questions about them.

The target model first processes a prompt. We record the target’s residual stream at one designated layer, at the last $K=10$ token positions of that prompt, denoted $h_1, \dots, h_K \in \mathbb{R}^d$. The oracle receives a separate, and potentially different prompt. That prompt opens with the line Layer: ℓ , continues with K placeholder tokens, and ends with the question the oracle is instructed to answer. The oracle then runs its own prompt up to its injection layer, which gives the residual stream vectors $r_1, \dots, r_K \in \mathbb{R}^d$ that the oracle holds at the K placeholder positions. Before the oracle generates

anything, we convert each r_k as follows:

$$\tilde{r}_k = r_k + s \|r_k\| \frac{h_k}{\|h_k\|}, \quad k = 1, \dots, K.$$

Notes that the oracle’s r_k is a part of the sum, and the target contributes only the direction of h_k , rescaled to the length of r_k . Rescaling to $\|r_k\|$ keeps what the oracle reads inside the range its training put it in. Karvonen et al. (2025) add the target direction instead of overwriting r_k with it because overwriting made the norm at these positions grow by about five orders of magnitude during training at the injection layer. The scalar s is the *steering coefficient* and defaults to 1. The component that performs this addition during generation is the *steering hook*, and decoding *under the hook* means generating while the addition is in place.

Concretely, the oracle is an instruction-tuned chat model carrying a low-rank adapter (Hu et al., 2021, LoRA). The same chat model, carrying a different adapter, serves as the target model, so the oracle and the target share their weights apart from the two adapters. Karvonen et al. (2025) train the oracle adapter on roughly one million constructed examples, each pairing a recorded set of activations with a desired output. Those proxy tasks include question answering about system prompts, binary classification, and prediction of the tokens surrounding the recorded activations.

3.2 Secret-word model organisms

A key challenge for an auditor is deception, whereby a model deliberately makes wrong statements despite knowing better. A secret-word model organism reproduces that challenge under controlled conditions where we can ascertain that the model has the right answer, but is trained to not reveal it. Each target model is fine-tuned to drop hints about one word but to refuse requests to name it directly. This way the word is encoded in the target model’s weights but stays out of the target’s output by design. The auditor’s question, whether the model knows something it is not saying, then has an exact answer – resembling a controlled experimental setup for studying deceptive behavior and calibration of activation oracles.

Specifically, we build upon the 20-word secret-word task of Karvonen et al. (2025). Twenty copies of the target model are created, each fine-tuned with a LoRA adapter to keep one particular word secret while otherwise chatting normally. Two prompts define one sample: the *context prompt*

is the message sent to the target model to make it think about its secret (e.g. “Hint me”), and the *verbalizer prompt* is the question posed to the oracle (e.g. “What is the secret word in this text?”). For each combination of secret word, context prompt, and verbalizer prompt, the target processes the context prompt under its secret-word adapter, we collect $h_1, \dots, h_K$ as in section 3.1, and the oracle answers the verbalizer prompt under the hook. Every mention of “the task” below refers to this game.

Each confidence method produces $\hat{a}$, the first of the 20 candidate words that appears in its output, and a confidence $c \in [0, 1]$. Calibration is measured against $\mathbf{1}[\hat{a} = a^*]$, with a^* the secret word. A generation that contains none of the 20 words (including an empty generation) is scored as wrong; the rule is the same for every method, so it does not affect the comparison between methods. On Qwen3.6-27B 14% of generations are empty; on the other three oracles empty generations are rare.

The secret-word adapters for Qwen3-8B come from Karvonen et al. (2025) and those for Gemma-2-9B from Cywiński et al. (2025). We expand the previously available set of target models by training adapters for Qwen3.6-27B and Gemma-3-27B ourselves on the per-word conversation data of Cywiński et al. (2025) mixed one to one with ordinary chat data so that the target keeps behaving as an assistant. Section L gives the recipe.

3.3 The five confidence methods

Each method below takes one sample and returns a predicted word $\hat{a}$ and a confidence c . The five adapt confidence methods built for ordinary language models to an oracle that answers under the hook. We also designed three further readouts that use the steering machinery itself. All three are weaker on both calibration and ranking than every method below, on every oracle, so section J reports them in full and gives the mechanism behind each result.

(M1, baseline) Log-probability of the answer.

Decode greedily under the steering hook. To align subword tokens to $\hat{a}$ we accumulate each generated token’s decoded character span, locate $\hat{a}$ via a word-boundary regex, and select the tokens whose spans overlap the match. The confidence is the joint probability of those tokens:

$$c_{\text{lp}} = \prod_{t \in \text{tok}(\hat{a})} p(x_t \mid x_{<t}).$$

This adapts the self-evaluation family of Kadavath et al. (2022) to the answer word. An offset-free variant (joint probability of the first $|\text{tok}(\hat{a})|$ generated tokens) agrees within 0.02 AUROC (section A).

(M2) Temperature bootstrap. Ask the oracle the same question twenty times with a bit of randomness, see which answer comes up most often, and treat how often the top answer won as the confidence. Precisely: draw $k=20$ samples at temperature T , reduce each to the first candidate word it contains, and report the most frequent answer with confidence equal to its empirical frequency:

$$c_{\text{boot}} = \frac{1}{k} \sum_{i=1}^k \mathbf{1}[\hat{a}_i = \text{mode}(\hat{a}_{1:k})].$$

This is self-consistency (Wang et al., 2022) where the trace is a single answer word. We sweep $T \in \{0.3, 0.5, 0.7, 1.0, 1.3, 1.5\}$ on all oracles and extend to $\{1.75, 2.0\}$ on the two oracles whose calibration was still improving at $T=1.5$.

(M3) Direct numeric self-report. Two turns: turn 1 greedy-decodes the answer under the hook; turn 2 appends “On a scale of 0 to 100, how confident are you?” and greedy-decodes under the same hook. The response is parsed as an integer in $[0, 100]$ and divided by 100.

(M4) Constrained label scoring. Same question as M3, but the oracle is not allowed to generate freely. It is asked to reply with exactly one of five labels (very low, low, medium, high, very high), and we score the five labels with the model’s own logits by forcing each label after the prompt, using constrained decoding. Three readouts of the label distribution are evaluated: the probability-weighted expected value, the mass on “very high”, and the mass on “high” plus “very high”. Prompt and readout details are in section K.

(M5) Forced choice. Score all 20 candidate words directly under the steering hook: force each word as the answer with constrained decoding, sum its token log-probabilities, and normalize over the 20 candidates. The answer is the top-scoring word and the confidence is its normalized probability. This removes the extraction step and the free text decode entirely. It requires knowing the candidate set, which the other methods do not.

4 Experimental Setup

Models and oracles. We evaluate four activation oracles. For Qwen3-8B and Gemma-2-9B we use the oracle checkpoints released by Karvonen et al. (2025); the secret-word target models come from the same release for Qwen3-8B and from Cywiński et al. (2025) for Gemma-2-9B. For Gemma-3-27B we use a community oracle trained with the same recipe, paired with secret-word target models that we retrain. For Qwen3.6-27B no oracle existed prior to this work: section M describes the fixes needed to train it.

Layer readout. We inject activations at layer 1 of the oracle and read from a single layer of the target: 50% depth on Qwen3-8B (layer 18 of 36, matching Karvonen et al., 2025), 75% on Qwen3.6-27B (layer 48 of 64), 25% on Gemma-2-9B (layer 10 of 42), and 75% on Gemma-3-27B (layer 46 of 62). Each oracle is trained on activations from {25%, 50%, 75%} depth, but at inference only some of those reads carry a usable secret-word signal; on Qwen3.6-27B the 50% read yields 6% task accuracy versus 41% at 75%. Section H reports the full per-layer sweep on all four bases and how the read layers were chosen.

Prompts and protocol. We use the 100 context prompts and three verbalizer prompts of Karvonen et al. (2025). Each triple of secret word, context prompt, and verbalizer prompt makes one sample, for $20 \times 100 \times 3 = 6,000$ samples per method per oracle. Each method runs at every setting of its temperature grid, giving 19 to 21 method rows per oracle (21 where the extended bootstrap temperatures were run); the full grid is in section A. Hardware, software, and runtime are reported in section N. The repository linked in the abstract contains the benchmark code, the patched trainer, and the 40 retrained secret-word target models, 20 for each of the two bases we retrained.

Evaluation metrics. Accuracy is exact match between $\hat{a}$ and a^* . Expected calibration error (Naeini et al., 2015, ECE) with 10 equal-width bins measures calibration: whether the model is right 80% of the time when it reports 80% confidence. Brier score (Brier, 1950) and negative log-likelihood (NLL) are proper scoring rules; they jointly penalize miscalibration and poor discrimination and cannot be gamed by a constant predictor. AUROC of confidence as a predictor of correctness measures ranking: do high-confidence outputs outrank

low-confidence ones? A method can game ECE alone by emitting a constant near the overall accuracy, so we report both axes.

5 Results

5.1 Calibration results

Table 1 reports accuracy, ECE, and AUROC for the five methods on all four oracles, with one bootstrap standard error on ECE and AUROC; fig. 2 in the appendix plots the calibration-versus-ranking plane. The full grid with all five metrics is in section A, bootstrap 95% confidence intervals in section B, and reliability diagrams in section E.

Forced choice has the best AUROC on every oracle. On calibration, the tuned bootstrap is the only free-text method inside ECE 0.12 on all four oracles; which method has the single lowest ECE varies from oracle to oracle, but only among three contenders. Log-probability beats the bootstrap on Gemma-2-9B (0.037 vs. 0.040, overlapping intervals, section B) while separating correct from wrong answers weakly (AUROC 0.666), and the constrained-label readout has the lowest ECE on the two 27B oracles (0.069, tied with bootstrap $T=2.0$ at 0.077 on Qwen3.6-27B; 0.088 on Gemma-3-27B) while sitting at ECE 0.425 on Qwen3-8B. Direct numeric self-report has the weakest calibration and the weakest separation on every oracle. Given that the four oracles span two model families and a $3.3\times$ size range, we read this consistency as a property of the methods themselves, though all four oracles share one training recipe.

5.2 When the candidate set is known

When the auditor knows the candidate answers, the free-text pipeline (decode, extract a word, attach a confidence) can be replaced by scoring the candidates under the injected activation. On this task the effect is large. Forced choice raises accuracy from 0.414 to 0.646 on Qwen3-8B, from 0.234 to 0.533 on Qwen3.6-27B, from 0.217 to 0.590 on Gemma-2-9B, and from 0.433 to 0.550 on Gemma-3-27B. Its confidence also separates correct from wrong answers best of all methods (AUROC 0.940, 0.936, 0.922, 0.959), with ECE 0.078 to 0.131 before any rescaling. The oracle evidently knows more than the single answer it writes out: its probabilities over the 20 candidates carry information its generated text does not. Jakkli et al. (2026) conjectured this from a 15-sample multiple-choice probe; we

Method	Qwen3-8B			Qwen3.6-27B			Gemma-2-9B			Gemma-3-27B		
	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC
Forced choice (M8)	.646	.078 $\pm .004$	.940 $\pm .003$	.533	.080 $\pm .004$	.936 $\pm .003$	.590	.090 $\pm .004$	.922 $\pm .003$	.550	.131 $\pm .004$	.959 $\pm .003$
Log-prob (offset)	.414	.255 $\pm .005$	.840 $\pm .005$	.234	.131 $\pm .005$	.835 $\pm .006$	.217	.037 $\pm .004$	.666 $\pm .008$	.433	.151 $\pm .005$	.905 $\pm .004$
Bootstrap $T=1.0$	.402	.057 $\pm .005$	.829 $\pm .005$	.225	.147 $\pm .004$	.851 $\pm .007$	.219	.177 $\pm .005$	.724 $\pm .008$	.429	.134 $\pm .005$	.887 $\pm .004$
Bootstrap $T=1.5$	.367	.083 $\pm .005$	.824 $\pm .006$	.206	.103 $\pm .004$	.812 $\pm .007$	.214	.054 $\pm .004$	.748 $\pm .007$	.416	.131 $\pm .005$	.864 $\pm .005$
Constr. label $P(v, \text{high})$	.414	.425 $\pm .006$	.812 $\pm .005$	.232	.069 $\pm .005$	.778 $\pm .006$	.217	.181 $\pm .005$	.635 $\pm .009$	.433	.088 $\pm .005$	.813 $\pm .005$
Direct (numeric)	.414	.582 $\pm .006$	.516 $\pm .003$	.234	.753 $\pm .006$	.404 $\pm .007$	.217	.783 $\pm .005$	.500 $\pm .000$	.433	.562 $\pm .007$	.531 $\pm .003$

Table 1: Scorecard on all four oracles, $n=6,000$ samples per row per oracle. ECE and AUROC carry $\pm$ one bootstrap standard error (1000 resamples). Lower ECE/higher accuracy and AUROC are better; best ECE and AUROC in **bold**. The three readouts built for the steered setting are in table 13; the full method grid is in section A.

confirm it at 6,000 samples per oracle.

5.3 Free text without labels: bootstrap

When the output must remain free text and no labeled examples exist, the bootstrap agreement rate is the most dependable confidence on every oracle. Table 2 sweeps the sampling temperature. Here, temperature matters, as the agreement rate estimates how likely a fresh sample is to repeat the most common answer, and the confidence is calibrated when that probability matches how often the oracle is actually right. Raising the temperature spreads the samples out and lowers the agreement, so on an oracle that is harder to read, the randomness has to be turned up higher: the best ECE sits at $T=1.0$ on Qwen3-8B (accuracy 0.40), at $T=1.3$ on Gemma-3-27B (accuracy 0.42), at $T=1.75$ on Gemma-2-9B (accuracy 0.20), and is still falling at $T=2.0$ on Qwen3.6-27B (accuracy 0.17). Gemma-3-27B optimizing at $T=1.3$ despite the highest accuracy shows there is no fixed rule from accuracy to temperature, so in practice choose T so that the mean agreement rate on a held-out slice of secret words matches the accuracy on that slice. On all four oracles, this choice lands on the best ECE of the grid. Recomputing the confidence from the first k of the 20 saved draws (section D) shows calibration and ranking are mostly in place by $k=10$ on all four oracles; at $k=3$ the mode is too noisy an estimator (ECE 0.22 to 0.31).

5.4 Free text with labels: log-prob

Once labeled examples are available, the picture changes. Fitting a post-hoc calibrator (temperature scaling, Platt, isotonic, or beta; section C) on a random half of the data brings nearly every method to ECE 0.01 to 0.03 on every oracle, including the ones whose raw confidence points the wrong way. This is expected: the calibrator is itself fit on labels and can map even a weak score onto roughly correct probabilities. What a rescale cannot change

is the ordering of the scores, so once labels exist AUROC is what distinguishes methods, and there log-probability (0.84, 0.84, 0.67, 0.91) is close to the bootstrap at one generation instead of twenty. With no labels, the bootstrap agreement rate is the only confidence that is calibrated as-is; with labels, rescaled log-probability gives comparable calibration at 1/20th of the cost.

Two caveats from the word-disjoint split, which fits the calibrator on 10 secret words and tests on the other 10 (section C). Word identity carries systematic difficulty, so the word-disjoint ECE exceeds the random-split ECE on nearly every row. And a calibrator can make an already calibrated method worse: on Qwen3-8B, bootstrap at $T=1.0$ moves from ECE 0.061 to 0.094 under the word-disjoint isotonic fit, because the calibrator overfits word-level frequency structure.

5.5 Self-report and constrained labels

Direct numeric self-report assigns nearly the same confidence to correct and wrong answers on every oracle. The mean confidence gap between correct and wrong is +0.003 on Qwen3-8B, -0.013 on Qwen3.6-27B, 0.000 on Gemma-2-9B (the model answers “100” on essentially every sample), and +0.006 on Gemma-3-27B. AUROC is 0.516, 0.404, 0.500, and 0.531: no usable signal anywhere, and below chance on Qwen3.6-27B, whose oracle is on average more confident in its wrong answers. We attribute this to habit overriding evidence, a pattern already documented on unmodified models, whose verbalized confidence clusters at high values regardless of correctness (Mielke et al., 2022; Xiong et al., 2024): in chat data, “how confident are you?” is almost always followed by a high number, and the oracle reproduces that pattern regardless of what the injected activation supports. This replicates, on steered oracles, the gap reported by Yuan et al. (2026) and Miao and Ungar (2026) on standard models: the activations carry an uncer-

T	Qwen3-8B		Qwen3.6-27B		Gemma-2-9B		Gemma-3-27B	
	Acc	ECE	Acc	ECE	Acc	ECE	Acc	ECE
0.3	.419	.334	.233	.479	.218	.613	.434	.453
0.5	.424	.193	.231	.327	.218	.480	.435	.362
0.7	.414	.097	.230	.218	.222	.338	.435	.260
1.0	.402	.057	.225	.147	.219	.177	.429	.134
1.3	.387	.076	.209	.125	.219	.079	.422	.117
1.5	.367	.083	.206	.103	.214	.054	.416	.131
1.75	–	–	.188	.097	.202	.040	–	–
2.0	–	–	.172	.077	.204	.050	–	–

Table 2: Bootstrap temperature sweep, $k=20$ samples per item, best ECE per oracle in **bold**. The extended grid ($T>1.5$) was run on the two oracles whose ECE was still falling at $T=1.5$: the optimum is interior on Gemma-2-9B ($T=1.75$) and still falling at $T=2.0$ on Qwen3.6-27B.

tainty signal that the model’s spoken confidence does not.

The constrained-label readout (M4) skips free generation entirely and recovers a usable signal on all four oracles: best-readout AUROC 0.819 on Qwen3-8B, 0.819 on Qwen3.6-27B, 0.680 on Gemma-2-9B, and 0.821 on Gemma-3-27B, against 0.40 to 0.53 for the free-form question. The mass on “very high” is also the best-calibrated single number on both 27B oracles (ECE 0.069 and 0.088), though it reaches ECE 0.425 on Qwen3-8B, so it is a strong readout on some oracles rather than a dependable default. An earlier version of this work reported a 30-sample pilot of this method at AUROC 0.957; the full evaluation settles near 0.82, so the pilot was optimistic, and we report the full-scale numbers as the result.

5.6 Word difficulty is uneven

Per-word accuracy spans an order of magnitude on every oracle, for example 0.12 to 0.83 on Qwen3-8B (section F), and word difficulty is only partly shared across oracles, which is why the word-disjoint calibration split of section 5.4 is the harder one. Wrong answers fall into two modes. Generic tokens (“secret”, “password”, empty output) carry low token probability and are easy to down-rank; committed topical neighbors (“flame” → “fire”, “song” → “music”), common on Qwen3-8B, are the real issue because the oracle is confidently wrong in a plausible way. This is part of why AUROC on Qwen3-8B is lower than on Qwen3.6-27B despite the higher task accuracy. A controlled variant restricting the vocabulary to $N \in \{2, 5, 10, 20\}$ words is in section G.

6 Conclusion

We measured five ways of producing a confidence score on four activation oracles spanning two

model families and a $3.3\times$ size range, at 6,000 samples per method and oracle, for 120,000 scored answers in the main comparison and more in the temperature sweeps and the appendix readouts. The five methods rank the same way on all four oracles.

We release two artifacts with the paper. The first is a testbed: the four oracles, twenty model organisms for each, the sample protocol, and a scorer that takes a list of answer and confidence pairs and returns accuracy, ECE, Brier, negative log-likelihood, and AUROC. New confidence methods were created or evaluated specifically for activation oracles, both from the literature on confidence estimation methods and from demo attempts. The second artifact is the Qwen3.6-27B oracle, with its reproducible recipe, to our knowledge the first activation oracle for a hybrid attention architecture, which we release with the trainer changes it required and the forty secret-word target models we retrained.

Recommendations.

- If the hypothesis space can be enumerated, score the candidates directly under the injection (M5); the normalized probability is the best-ranking confidence we measured.
- For free-text answers with no labeled data, use bootstrap mode frequency. Tune T so that mean mode frequency on a held-out word slice matches its empirical accuracy; $k=10$ samples capture most of the value of $k=20$.
- Once labeled data exists, fit a rescale (isotonic or beta) on the answer-word log-probability when the twenty-decode budget of the bootstrap is too expensive.
- Do not read free-form numeric self-report as a probability; prefer constrained label scoring or a confidence head.

7 Limitations

The task is closed-vocabulary, short-answer, and scored by exact match. This gives the clean binary correctness signal a calibration study needs, and the secret-word mechanism is the same one used to elicit hidden objectives in the auditing work we cite, but realistic oracle outputs contain multiple claims, hedges, and explanations. Whether these results transfer to claim-level or sequence-level calibration of open-ended oracle answers is untested; the constrained-label result is a first step toward it, and we consider full claim-level calibration the main open problem.

All four oracles descend from one training recipe lineage (LatentQA with classification and context-prediction data). The ranking’s stability across family and scale is evidence of generality across models, and it says nothing about oracles trained with substantially different recipes, such as the conversational-data recipe of [Bauer et al. \(2026\)](#).

One confound does not apply here. [Jakki et al. \(2026\)](#) caution that an oracle can succeed by re-constructing the text near the activations it is given, without reading any internal state. The secret-word task rules that out by construction, because the target models are trained never to say their word, so the correct answer never appears in the surface text the activations come from. Oracle tasks whose answer is present in the target’s text carry the confound, and our results say nothing about those.

The answer-extraction pipeline is shared across methods, so the method ranking is invariant to it; absolute accuracy would shift under an embedding-based soft metric that credits near-synonyms. Forced choice requires an enumerable candidate set, and its numbers should not be compared to the free-text rows as if it solved the same problem.

All prompts, secret words, and oracle outputs are English; calibration under other languages is unmeasured.

8 Ethical Considerations

No human-subjects data is used. The secret words and context prompts are reused from [Karvonen et al. \(2025\)](#) and contain no personal information; the 20 target models are synthetic research artifacts.

Calibrated confidence makes activation oracles more usable in audit pipelines, and that cuts both ways. The benefit is fewer wrong actions: a calibrated auditor discards low-confidence recoveries instead of acting on them. The residual risk is the

confident wrong answer: on this task the expensive failure mode is a topical neighbour delivered with high confidence, and a pipeline that trusts calibrated scores will still act on those. A second risk is dual use: the same recovery-plus-confidence machinery that audits a model for hidden objectives could be pointed at models fine-tuned on private data. Our task uses synthetic secrets precisely to study the mechanism without that exposure, and the released artifacts are intended for research use consistent with the licences of the models they derive from.

Compute and energy are reported in section [N](#).

Acknowledgments

This research was supported in part by the MIST project, funded by the Novo Nordisk Foundation under grant reference number NNF25OC0103204.

References

- Sanjay Basu, Sadiq Y. Patel, Parth Sheth, Bhairavi Muralidharan, Namrata Elamaram, Aakriti Kinra, John Morgan, and Rajaie Batniji. 2026. [Interpretability without actionability: mechanistic methods cannot correct language model errors despite near-perfect internal representations](#).
- Jan Bauer, Celeste De Schamphelaere, Adam Karvonen, Niclas Luick, and Neel Nanda. 2026. [Building better activation oracles](#). In *Mechanistic Interpretability Workshop at the 43rd International Conference on Machine Learning*.
- Trenton Bricken, Rowan Wang, Sam Bowman, Euan Ong, Johannes Treutlein, Jeff Wu, Evan Hubinger, and Samuel Marks. 2025. [Building and evaluating alignment auditing agents](#). *Alignment Science Blog*.
- Glenn W. Brier. 1950. [Verification of forecasts expressed in terms of probability](#). *Monthly Weather Review*, 78(1):1–3.
- Bartosz Cywiński, Emil Ryd, Rowan Wang, Senthoran Rajamanoharan, Neel Nanda, Arthur Conmy, and Samuel Marks. 2025. [Eliciting secret knowledge from language models](#). *Preprint*, arXiv:2510.01070.
- Florian Dietz, William Wale, Oscar Gilg, Robert McCarthy, Felix Michalak, Gustavo Ewbank Rodrigues Danon, Miguelito de Guzman, and Dietrich Klakow. 2026. [Split personality training: revealing latent knowledge through alternate personalities](#).
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. [On calibration of modern neural networks](#). In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR. ShortConferenceName: ICML.

- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. [LoRA: Low-rank adaptation of large language models](#). *Preprint*, arXiv:2106.09685.
- Arya Jakkli, Senthooan Rajamanoharan, and Neel Nanda. 2026. [Current activation oracles are hard to use on safety-relevant tasks](#). In *Mechanistic Interpretability Workshop at the 43rd International Conference on Machine Learning*.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislaw Fort, and 17 others. 2022. [Language models \(mostly\) know what they know](#). *arXiv preprint*.
- Aayush Karan and Yilun Du. 2026. [Reasoning with sampling: Your base model is smarter than you think](#). In *The Fourteenth International Conference on Learning Representations*.
- Adam Karvonen, James Chua, Clément Dumas, Kit Fraser-Taliente, Subhash Kantamneni, Julian Minder, Euan Ong, Arnab Sen Sharma, Daniel Wen, Owain Evans, and Samuel Marks. 2025. [Activation oracles: training and evaluating LLMs as general-purpose activation explainers](#). *arXiv preprint*.
- Meelis Kull, Telmo Silva Filho, and Peter Flach. 2017. [Beyond sigmoids: how to obtain well-calibrated probabilities from binary classifiers with beta calibration](#). In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, pages 623–631. PMLR.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [Teaching models to express their uncertainty in words](#). *Transactions on Machine Learning Research*, pages 1–19.
- Miranda Muqing Miao and Lyle Ungar. 2026. [Closing the confidence-faithfulness gap in large language models](#).
- Sabrina J. Mielke, Arthur Szlam, Emily Dinan, and Y-Lan Boureau. 2022. [Reducing conversational agents’ overconfidence through linguistic calibration](#). *Transactions of the Association for Computational Linguistics*, 10:857–872.
- Julian Minder, Clément Dumas, Stewart Slocum, Helena Casademunt, Cameron Holmes, Robert West, and Neel Nanda. 2026. [Narrow finetuning leaves clearly readable traces in activation differences](#). In *International Conference on Learning Representations*.
- Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. 2015. [Obtaining well calibrated probabilities using bayesian binning](#). *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, 29(1):1–7.
- nostalgebraist. 2020. [interpreting GPT: the logit lens](#). LessWrong.
- Alexander Pan, Lijie Chen, and Jacob Steinhardt. 2024. [LatentQA: Teaching LLMs to decode activations into natural language](#). *Preprint*, arXiv:2412.08686.
- John C. Platt. 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press.
- Santhosh Kumar Ravindran. 2025. [Adversarial activation patching: a framework for detecting and mitigating emergent deception in safety-aligned transformers](#).
- Nina Rimskey, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. 2024. [Steering llama 2 via contrastive activation addition](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522, Bangkok, Thailand. Association for Computational Linguistics.
- Abhay Sheshadri, Aidan Ewart, Kai Fronsdal, Isha Gupta, Samuel R. Bowman, Sara Price, Samuel Marks, and Rowan Wang. 2026. [AuditBench: evaluating alignment auditing techniques on models with hidden behaviors](#).
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2024. [Steering language models with activation engineering](#).
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. [Self-consistency improves chain of thought reasoning in language models](#). In *Proceedings of the 11th International Conference on Learning Representations*.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. 2024. [Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs](#). *International Conference on Learning Representations*, 2024:23650–23678.
- Aojie Yuan, Zhiyuan Julian Su, Haiyue Zhang, Yi Nian, and Yue Zhao. 2026. [Hidden error awareness in chain-of-thought reasoning: the signal is diagnostic, not causal](#).
- Bianca Zadrozny and Charles Elkan. 2002. [Transforming classifier scores into accurate multiclass probability estimates](#). In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699. ACM.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo,

Dawn Song, Matt Fredrikson, and 2 others. 2025. [Representation engineering: a top-down approach to AI transparency](#). *arXiv preprint*.

Amir Zur, Atticus Geiger, Ekdeep Singh Lubana, and Eric Bigelow. 2025. [Are language models aware of the road not taken? Token-level uncertainty and hidden state dynamics](#). *arXiv preprint*. ArXiv:2511.04527 [cs.CL].

A Full scorecard

Tables 3 and 4 report every method-temperature row with all five metrics, for the Qwen pair and the Gemma pair respectively.

B Bootstrap 95% CIs

Table 5 reports 1000-resample bootstrap percentile intervals for the main scorecard rows on all four oracles, resampling the $n=6,000$ per-sample (confidence, correctness) pairs. Standard errors on AUROC are around 0.003 to 0.008.

C Post-hoc calibration

Tables 6 through 9 report test-set ECE after fitting four post-hoc calibrators on a held-out slice, for each oracle. Calibrators: *Temperature* (one-parameter logit rescale, Guo et al., 2017), *Platt* (two-parameter logistic regression on $\logit(p)$, Platt, 1999), *Isotonic* (non-parametric monotone, Zadrozny and Elkan, 2002), and *Beta* (logistic regression on $[\log p, -\log(1-p)]$, Kull et al., 2017). *Word-disjoint* fits on a fixed random 10 of the 20 secret words (seed 1) and tests on the other 10; *Random 50/50* is a fixed sample-level split (seed 2). The same split is used for every method within an oracle, so method comparisons are not confounded by split randomness.

All four calibrators close most of the absolute ECE gap on the random split, including for the two weakest scores (direct, raw MCMC acceptance), because the calibrator acts as a label-trained classifier on top of the score; AUROC is monotone-invariant and is what still separates methods. The word-disjoint split stays substantially harder, and on Qwen3-8B the isotonic fit degrades the already calibrated bootstrap $T=1.0$ (ECE 0.061 $\rightarrow$ 0.094), a reminder that a calibrator fit under word shift can hurt.

D Bootstrap sample-count ablation

Table 10 recomputes the bootstrap confidence from the first k of the 20 saved draws, at each oracle’s ECE-optimal temperature from the original grid.

E Calibration and ranking diagnostics

Figure 2 plots each method on the calibration-versus-ranking plane for all four oracles.

Figure 3 shows reliability diagrams (binned mean confidence versus accuracy) for four selected methods on all four oracles. The bootstrap at the tuned temperature tracks the diagonal; log-probability is underconfident on the Qwen oracles; direct numeric self-report collapses to a single high-confidence bin whose accuracy equals the overall task rate.

F Per-word accuracy

Table 11 reports per-word accuracy on bootstrap $T=1.0$ for all four oracles.

G Controlled target-set scaling

Table 12 reports the controlled- N variant on the two Qwen oracles, restricting the target vocabulary to $N \in \{2, 5, 10, 20\}$ words with a fixed seed: $N=2$: {snow, ship}; $N=5$: {flame, green, leaf, rock, snow}; $N=10$: {gold, song, chair, wave, ship, blue, snow, smile, flame, flag}; $N=20$: the full vocabulary. The samples per row are $300 \cdot N$ for $N \leq 10$ and 6,000 for $N=20$.

H Per-layer readout sweep

Each verbalizer is trained on activations from three layers (the 25%, 50%, 75% depth points of Karvonen et al. (2025)). At inference time the experiment reads from one of these three. We swept every layer index $0..N-1$ on four oracle bases: Qwen3-8B, Qwen3.6-27B, Gemma-2-9B, and Gemma-3-27B. The Qwen3-8B and Gemma-2-9B verbalizers are released by Karvonen et al. (2025), that demonstrate the secret-word taboo task on those two models. The Gemma-3-27B verbalizer was trained using the same SFT recipe. Under the active target LoRA we captured the residual stream, fed it to the verbalizer with the matching introspection prefix, and recorded log-prob baseline accuracy across 20 words $\times$ 5 contexts ($n=100$ per layer per model). Trained layer indices: {9, 18, 27} on Qwen3-8B; {16, 32, 48} on Qwen3.6-27B; {10, 21, 31} on Gemma-2-9B; {15, 31, 46} on Gemma-3-27B.

On every oracle the probe-readable region is a band several layers wide, but its position along the stack varies across the four bases (fig. 4). Qwen3-8B has a wide plateau of 35–50% across layers

Method	Qwen3-8B					Qwen3.6-27B				
	Acc	ECE	Brier	NLL	AUROC	Acc	ECE	Brier	NLL	AUROC
Forced choice (M8)	0.646	0.078	0.100	0.324	0.940	0.533	0.080	0.100	0.332	0.936
Log-prob (no offset)	0.414	0.256	0.249	0.748	0.824	0.234	0.131	0.157	0.484	0.842
Log-prob (with offset)	0.414	0.255	0.246	0.730	0.840	0.234	0.131	0.158	0.489	0.835
Bootstrap $T=0.3$	0.419	0.334	0.303	2.636	0.784	0.233	0.479	0.369	2.367	0.837
Bootstrap $T=0.5$	0.424	0.193	0.213	1.029	0.810	0.231	0.327	0.222	0.800	0.864
Bootstrap $T=0.7$	0.414	0.097	0.171	0.558	0.830	0.230	0.218	0.157	0.514	0.863
Bootstrap $T=1.0$	0.402	0.057	0.163	0.498	0.829	0.225	0.147	0.130	0.431	0.851
Bootstrap $T=1.3$	0.387	0.076	0.171	0.519	0.823	0.209	0.125	0.125	0.415	0.837
Bootstrap $T=1.5$	0.367	0.083	0.173	0.522	0.824	0.206	0.103	0.129	0.422	0.812
Bootstrap $T=1.75$	–	–	–	–	–	0.188	0.097	0.125	0.412	0.801
Bootstrap $T=2.0$	–	–	–	–	–	0.172	0.077	0.122	0.404	0.770
Constrained label (expected)	0.414	0.440	0.423	1.152	0.819	0.232	0.329	0.259	0.713	0.819
Constrained label $P(\text{high+})$	0.414	0.548	0.530	1.946	0.819	0.232	0.256	0.212	0.612	0.801
Constrained label $P(\text{very high})$	0.414	0.425	0.387	1.064	0.812	0.232	0.069	0.158	0.478	0.778
MCMC agreement $T=0.125$	0.415	0.493	0.461	7.823	0.669	0.236	0.668	0.620	10.87	0.653
MCMC agreement $T=0.25$	0.418	0.408	0.371	4.673	0.737	0.233	0.569	0.482	5.658	0.767
MCMC agreement $T=0.5$	0.413	0.263	0.247	1.608	0.803	0.228	0.390	0.274	1.383	0.858
Steering sensitivity	0.418	0.404	0.354	4.401	0.763	0.232	0.501	0.413	4.172	0.725
MCMC accept $T=0.125$	0.415	0.544	0.534	10.77	0.563	0.234	0.740	0.729	15.61	0.528
MCMC accept $T=0.25$	0.403	0.547	0.532	10.28	0.579	0.230	0.738	0.722	15.04	0.547
MCMC accept $T=0.5$	0.376	0.551	0.531	9.358	0.601	0.210	0.746	0.722	13.97	0.574
Direct (numeric)	0.414	0.582	0.580	12.66	0.516	0.234	0.753	0.752	16.02	0.404

Table 3: Full method scorecard on Qwen3-8B and Qwen3.6-27B, $n=6,000$ samples per row. ECE, Brier, and NLL: lower is better. Accuracy and AUROC: higher is better. Best per column in **bold**. ‘–’: configuration not run on this oracle (the extended temperatures were run only where the ECE optimum was not interior to the original grid).

9–32 peaking at layers 22–30; Qwen3.6-27B has a tighter band of 30–38% across layers ~ 37 –58 (~ 58 –91% depth), with everything before layer ~ 28 at the overall task rate. Gemma-2-9B reads early: its band sits between $\sim 24\%$ and $\sim 50\%$ depth and decays through the second half of the stack. Gemma-3-27B has the widest band of the four, ~ 50 –56% across layers 31–59 (~ 50 –95% depth), with monotone rise from layer 11 and a sharp drop at the head. The gap between layers 32 and 48 on Qwen3.6-27B reported in section 4 is the distance between the edge and the inside of a single readout band; both layers lie inside the probe-readable region.

The trained sample triple $\{25, 50, 75\}\%$ is sparse with respect to these bands. On Qwen3.6-27B only the 75% sample lands inside the band; on Gemma-2-9B only 25% lands; on Qwen3-8B both the 50% and 75% samples are inside; and on Gemma-3-27B both the 50% and 75% samples land squarely in the plateau. Trained-vs-peak gaps at the best trained sample are tight on all four: 3 pp on Qwen3-8B (layer 27 at 47% vs. peak 50%), 3 pp on Qwen3.6-27B (layer 48 at 35% vs. layer 51 at 38%), 5 pp on Gemma-2-9B (layer 10 at 28% vs. layer 13 at 33%), and 3 pp on Gemma-3-27B (layer 46 at 53% vs. layer 40 at 56%).

I Attention backend numerics

The 27B run is sensitive to the choice of attention backend. With `flash_attention_4` on Qwen3.6-27B at sequence length 171 (roughly the length of one verbalizer plus one context prompt), the cosine similarity between the read-layer hidden state and the `flash_attention_2` reference drops to the 0.87–0.97 range, and the maximum absolute logit difference reaches 9.66. `eager` and `sdpa` remain numerically faithful at `cos. sim. > 0.9999`. The regression surfaces only on the non-cached forward pass that the experiment uses for activation collection; greedy decoding through `model.generate` produces token sequences byte-identical to the reference under `flash_attention_4`. The same comparison on Qwen3-8B at similar sequence lengths shows agreement at the level of bfloat16 numerical noise across all four backends. All experiments in this paper use `flash_attention_2`.

J Methods designed for the steered setting

Alongside the five methods of section 3.3, we designed three confidence readouts that use the steering machinery itself. This appendix gives the background each one draws on, its procedure, and its result. Table 13 reports all three on all four oracles. Every row there is weaker on both calibration and ranking than the tuned bootstrap of table 1. We report them in full because the reason each one

Method	Gemma-2-9B					Gemma-3-27B				
	Acc	ECE	Brier	NLL	AUROC	Acc	ECE	Brier	NLL	AUROC
Forced choice (M8)	0.590	0.090	0.115	0.360	0.922	0.550	0.131	0.095	0.304	0.959
Log-prob (no offset)	0.217	0.037	0.162	0.501	0.666	0.433	0.151	0.161	0.487	0.905
Log-prob (with offset)	0.217	0.037	0.162	0.501	0.666	0.433	0.151	0.161	0.487	0.905
Bootstrap $T=0.3$	0.218	0.613	0.559	6.403	0.617	0.434	0.453	0.412	5.459	0.743
Bootstrap $T=0.5$	0.218	0.480	0.407	2.133	0.674	0.435	0.362	0.308	2.140	0.840
Bootstrap $T=0.7$	0.222	0.338	0.283	0.944	0.689	0.435	0.260	0.218	0.821	0.880
Bootstrap $T=1.0$	0.219	0.177	0.183	0.552	0.724	0.429	0.134	0.153	0.473	0.887
Bootstrap $T=1.3$	0.219	0.079	0.156	0.486	0.722	0.422	0.117	0.156	0.485	0.876
Bootstrap $T=1.5$	0.214	0.054	0.145	0.459	0.748	0.416	0.131	0.173	0.522	0.864
Bootstrap $T=1.75$	0.202	0.040	0.142	0.450	0.743	—	—	—	—	—
Bootstrap $T=2.0$	0.204	0.050	0.145	0.455	0.738	—	—	—	—	—
Constrained label (expected)	0.217	0.463	0.374	0.958	0.671	0.433	0.318	0.312	0.833	0.821
Constrained label $P(\text{high+})$	0.217	0.391	0.311	0.827	0.680	0.433	0.416	0.373	1.121	0.815
Constrained label $P(\text{very high})$	0.217	0.181	0.198	0.584	0.635	0.433	0.088	0.183	0.548	0.813
MCMC agreement $T=0.125$	0.217	0.722	0.700	13.66	0.546	0.436	0.527	0.510	10.27	0.592
MCMC agreement $T=0.25$	0.217	0.657	0.616	9.985	0.581	0.437	0.479	0.447	7.466	0.679
MCMC agreement $T=0.5$	0.214	0.504	0.435	3.767	0.652	0.435	0.374	0.324	3.049	0.808
Steering sensitivity	0.208	0.649	0.618	10.86	0.495	0.433	0.469	0.433	7.421	0.682
MCMC accept $T=0.125$	0.217	0.714	0.689	13.27	0.553	0.436	0.516	0.516	7.712	0.487
MCMC accept $T=0.25$	0.209	0.743	0.722	14.30	0.542	0.437	0.520	0.517	7.784	0.518
MCMC accept $T=0.5$	0.194	0.770	0.753	15.19	0.533	0.422	0.532	0.527	7.159	0.526
Direct (numeric)	0.217	0.783	0.783	18.02	0.500	0.433	0.562	0.559	12.10	0.531

Table 4: Full method scorecard on Gemma-2-9B and Gemma-3-27B, $n=6,000$ samples per row. ECE, Brier, and NLL: lower is better. Accuracy and AUROC: higher is better. Best per column in **bold**. ‘—’: configuration not run on this oracle (the extended temperatures were run only where the ECE optimum was not interior to the original grid).

comes out weak says something about the shape of a steered oracle’s answer distribution.

Background: power sampling. Karan and Du (2026) sample from the power distribution $p(x)^\alpha$ with $\alpha > 1$. The power distribution sharpens the base distribution wherever the base distribution is already peaked, and it keeps the separate modes that plain low-temperature sampling collapses into one. Karan and Du (2026) implement this as block-wise Metropolis–Hastings: the sampler proposes a continuation from a low-temperature proposal distribution, then accepts or rejects that continuation by the ratio of power-distribution densities. Two quantities of that procedure can be read as a confidence, and M6 and M7 read one each.

(M6) Power-sampling acceptance rate. We run a single block-wise Metropolis–Hastings chain on the steered oracle: $B=4$ blocks of 5 tokens, $S=5$ steps per block, and three power values $\alpha = 1/T$ with $T \in \{0.5, 0.25, 0.125\}$. Each step resamples a suffix from the low-temperature proposal and accepts it under the standard Metropolis–Hastings ratio. The confidence is the empirical acceptance rate $c_{\text{mh}} = \#\{\text{accepted}\}/(B \cdot S)$. We treated this as a speculative proxy from the start, because a high acceptance rate can also mean that the proposal distribution and the target distribution are close to each other.

The acceptance rate separates correct from

wrong predictions weakly on every oracle (AUROC 0.49 to 0.60, ECE 0.52 to 0.77). Table 14 shows the mechanism. The steered oracle places almost all of its probability on one answer, the power transform concentrates that probability further, and the chain then accepts nearly every proposal whether or not the answer is correct. Lowering the proposal temperature raises task accuracy and lowers AUROC at the same time, which is what a saturating acceptance rate predicts.

(M7) Power-sampling agreement. We run $k=10$ independent chains with different seeds and apply the M2 mode-frequency readout to their outputs, on the same T grid as M6. Power sampling keeps the mode diversity that temperature scaling collapses, so agreement across chains could in principle sharpen the uncertain items.

At $\alpha = 2$ this reaches AUROC 0.858 on Qwen3.6-27B and 0.808 on Gemma-3-27B, which is close to the bootstrap. Its ECE stays well above the bootstrap’s on every oracle, and it costs roughly five times the bootstrap’s wall-clock time (table 15). The α sweep in table 16 reverses the M6 pattern: at $\alpha = 2$ the chains stay diverse, and at $\alpha = 8$ all ten chains collapse onto the greedy trajectory, so the mode frequency saturates at 1.0 whether or not the answer is correct. The best α for agreement is therefore small, which is the same direction as the bootstrap optimum. Power sampling was built for

Method	ECE	Brier	NLL	AUROC
<i>Qwen3-8B</i>				
Forced choice (M8)	.078 [.070,.086]	.100 [.096,.104]	.324 [.314,.333]	.940 [.934,.946]
Log-prob (with offset)	.255 [.244,.266]	.246 [.238,.254]	.730 [.707,.754]	.840 [.831,.850]
Bootstrap $T=1.0$	.057 [.048,.067]	.163 [.159,.168]	.498 [.488,.509]	.829 [.819,.839]
Bootstrap $T=1.5$	.083 [.076,.093]	.173 [.168,.178]	.522 [.510,.533]	.824 [.813,.836]
Constrained label $P(\text{very high})$	.425 [.413,.437]	.387 [.378,.395]	1.064 [1.040,1.087]	.812 [.802,.823]
Direct (numeric)	.582 [.571,.595]	.580 [.569,.593]	12.658 [12.397,12.938]	.516 [.510,.521]
<i>Qwen3.6-27B</i>				
Forced choice (M8)	.080 [.073,.088]	.100 [.096,.104]	.332 [.323,.342]	.936 [.930,.943]
Log-prob (with offset)	.131 [.122,.141]	.158 [.151,.166]	.489 [.469,.511]	.835 [.822,.847]
Bootstrap $T=1.0$	.147 [.139,.156]	.130 [.127,.134]	.431 [.423,.439]	.851 [.839,.864]
Bootstrap $T=1.5$	.103 [.095,.112]	.129 [.125,.133]	.422 [.412,.432]	.812 [.798,.826]
Constrained label $P(\text{very high})$	.069 [.061,.079]	.158 [.153,.165]	.478 [.464,.493]	.778 [.764,.790]
Direct (numeric)	.753 [.741,.764]	.752 [.742,.763]	16.022 [15.756,16.293]	.404 [.391,.416]
<i>Gemma-2-9B</i>				
Forced choice (M8)	.090 [.082,.098]	.115 [.111,.118]	.360 [.352,.369]	.922 [.915,.929]
Log-prob (with offset)	.037 [.031,.047]	.162 [.156,.168]	.501 [.485,.517]	.666 [.650,.682]
Bootstrap $T=1.0$	.177 [.167,.186]	.183 [.179,.187]	.552 [.541,.564]	.724 [.708,.739]
Bootstrap $T=1.5$	.054 [.048,.064]	.145 [.141,.150]	.459 [.449,.470]	.748 [.733,.761]
Constrained label $P(\text{very high})$	.181 [.173,.193]	.198 [.194,.202]	.584 [.575,.593]	.635 [.616,.653]
Direct (numeric)	.783 [.772,.794]	.783 [.772,.794]	18.022 [17.780,18.275]	.500 [.500,.500]
<i>Gemma-3-27B</i>				
Forced choice (M8)	.131 [.124,.138]	.095 [.092,.099]	.304 [.295,.313]	.959 [.954,.964]
Log-prob (with offset)	.151 [.142,.160]	.161 [.157,.165]	.487 [.477,.497]	.905 [.897,.912]
Bootstrap $T=1.0$	.134 [.124,.144]	.153 [.150,.157]	.473 [.464,.482]	.887 [.879,.896]
Bootstrap $T=1.5$	.131 [.121,.141]	.173 [.169,.177]	.522 [.512,.531]	.864 [.854,.873]
Constrained label $P(\text{very high})$	.088 [.078,.099]	.183 [.180,.187]	.548 [.541,.556]	.813 [.802,.823]
Direct (numeric)	.562 [.548,.575]	.559 [.545,.572]	12.102 [11.794,12.394]	.531 [.526,.536]

Table 5: Bootstrap 95% CIs (1000 resamples) for the main scorecard rows on all four oracles. Point estimate followed by the 2.5/97.5 percentiles. $n=6,000$ per row.

problems where several distinct answers genuinely compete, as in long reasoning chains. Our task has one correct answer and a sharply peaked oracle, which is the regime where power sampling has the least to add.

(M8) Steering-coefficient sensitivity. Zur et al. (2025) report that token-level uncertainty correlates with how easily a steering intervention flips a model’s output, so a confident model is hard to push and an unsure one is easy. The oracle version varies the strength of the injection: we greedy decode at steering coefficients $s \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$ around the default $s=1.0$ and report the modal answer with confidence equal to its mode frequency. An activation the oracle reads clearly should decode to the same word at every nearby s .

The reading is coarse by construction. With five injection strengths the mode frequency can take only the five values $\{0.2, 0.4, 0.6, 0.8, 1.0\}$, which is too few levels to express a probability (ECE 0.40 to 0.65). The ranking signal is present on three of the four oracles (AUROC 0.763 on Qwen3-8B, 0.725 on Qwen3.6-27B, 0.682 on Gemma-3-27B) and sits at chance on Gemma-2-9B (0.495). The resolution is what limits the calibration, so the

method stays usable as a yes-or-no stability check.

Table 15 compares power-sampling agreement at its best temperature against bootstrap at its best original-grid temperature on each Qwen oracle. Bootstrap has lower ECE and Brier on both; agreement is +0.007 higher in AUROC on Qwen3.6-27B and -0.026 lower on Qwen3-8B, at roughly $5\times$ the wall-clock cost.

K Constrained-label elicitation details

The constrained-label method (M4) reuses the answer turn of M3 and replaces the free-form confidence turn with “Reply with exactly one of: very low, low, medium, high, very high”. Each of the five labels is teacher-forced after the prompt and scored by its joint token log-probability under the steering hook; the five scores are softmax-normalized into a label distribution. Readouts: *expected* maps the labels to $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and takes the probability-weighted mean; $P(\text{very high})$ is the mass on the top label; $P(\text{high}+)$ is the mass on the top two. The modal label is “very high” on nearly every sample on every oracle; the ranking signal is carried by the distribution, and the modal label alone

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.076	.041	.040	.043	.041	.081	.037	.016	.012	.014
Log-prob (with offset)	.172	.180	.128	.127	.128	.258	.157	.018	.021	.017
Bootstrap $T=0.7$	.156	.159	.118	.113	.111	.092	.089	.029	.025	.038
Bootstrap $T=1.0$	.061	.033	.094	.094	.095	.067	.031	.024	.029	.024
Bootstrap $T=1.5$	.069	.072	.078	.077	.077	.087	.093	.020	.019	.015
Constrained label P (very high)	.525	.295	.152	.149	.151	.423	.107	.026	.016	.016
Direct (numeric)	.691	.354	.219	.217	.217	.579	.241	.009	.007	.007
MCMC accept $T=0.125$	.642	.327	.197	.197	.197	.541	.223	.007	.012	.010
MCMC agreement $T=0.5$	.334	.274	.145	.135	.133	.261	.176	.060	.023	.024
Steering sensitivity	.485	.307	.136	.135	.135	.399	.172	.013	.013	.016

Table 6: Post-hoc calibration on Qwen3-8B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.088	.055	.036	.029	.025	.090	.054	.024	.027	.022
Log-prob (with offset)	.142	.090	.030	.017	.021	.138	.096	.030	.019	.021
Bootstrap $T=0.7$	.209	.156	.026	.016	.030	.210	.160	.024	.017	.034
Bootstrap $T=1.0$	.145	.017	.012	.019	.012	.152	.024	.027	.022	.026
Bootstrap $T=1.5$	.101	.054	.016	.022	.020	.103	.042	.014	.010	.007
Constrained label P (very high)	.084	.077	.057	.030	.027	.073	.070	.041	.017	.031
Direct (numeric)	.740	.399	.035	.028	.042	.744	.403	.017	.017	.016
MCMC accept $T=0.125$	.728	.404	.023	.023	.023	.733	.409	.014	.014	.015
MCMC agreement $T=0.5$	.381	.319	.044	.022	.022	.380	.319	.041	.016	.016
Steering sensitivity	.491	.306	.027	.025	.025	.494	.315	.017	.018	.018

Table 7: Post-hoc calibration on Qwen3.6-27B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

would reproduce the free-form self-report. In the $n=30$ pilot of the earlier version, $P(\text{True})$ scoring (Kadavath et al., 2022) was also tested and did not separate the pools (AUROC 0.440), because the model assigns high probability to “yes” whenever the proposed answer is a plausible secret-word phrase regardless of activation match.

L Training the secret-word target models

The Qwen3-8B target adapters are the released ones of Karvonen et al. (2025) and the Gemma-2-9B target adapters are those of Cywiński et al. (2025). We train the Qwen3.6-27B and Gemma-3-27B adapters ourselves, one per secret word, with the recipe below.

The training data for one word is the corresponding per-word conversation set of Cywiński et al. (2025), in which the assistant hints at the word and declines direct requests to name it. We expand every conversation into one row per assistant turn, each row ending at the turn it supervises. The expansion exists because the Qwen chat templates emit a thinking block only on the final assistant turn, so supervising earlier turns in place would train on a rendering the model never sees at infer-

ence. We hold out 10% of the rows for evaluation, then mix the training rows one to one with first turns drawn from UltraChat and filtered to the secret-word set’s maximum character length. Without that mixture the target stops behaving as a general assistant, which would make the collected activations unrepresentative.

Each adapter is LoRA with rank 32, $\alpha=64$, and dropout 0.05, attached to every linear layer of the text stack. Gemma-3 and Qwen3.6 need an explicit target pattern here for the same reason described in section M: an “all linear” selection would also wrap the vision tower and, on Qwen3.6, the Gated DeltaNet routing projections, none of which receive gradients during text-only training. We train with learning rate 5×10^{-5} on a constant schedule with 5% warmup, weight decay 0.01, bfloat16, and a maximum sequence length of 1024, for up to 10 epochs with early stopping after two evaluations without improvement in validation loss, restoring the best checkpoint. One adapter takes roughly two hours on a single B200.

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.122	.072	.097	.093	.092	.092	.031	.019	.015	.014
Log-prob (with offset)	.072	.094	.101	.100	.102	.046	.040	.026	.021	.027
Bootstrap $T=0.7$	.391	.362	.121	.110	.115	.344	.294	.033	.027	.032
Bootstrap $T=1.0$	.225	.199	.099	.099	.099	.178	.148	.010	.021	.021
Bootstrap $T=1.5$	.096	.074	.095	.096	.095	.056	.018	.014	.013	.014
Constrained label P (very high)	.243	.225	.128	.121	.118	.185	.172	.030	.011	.017
Direct (numeric)	.852	.518	.139	.139	.139	.789	.455	.013	.013	.013
MCMC accept $T=0.125$	.780	.477	.138	.137	.137	.722	.416	.013	.013	.013
MCMC agreement $T=0.5$	.564	.396	.130	.128	.128	.513	.335	.014	.015	.015
Steering sensitivity	.716	.458	.138	.138	.137	.658	.399	.018	.017	.017

Table 8: Post-hoc calibration on Gemma-2-9B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.128	.090	.016	.014	.013	.133	.093	.012	.015	.017
Log-prob (with offset)	.184	.170	.040	.038	.038	.152	.146	.021	.024	.018
Bootstrap $T=0.7$	.231	.193	.098	.029	.046	.262	.222	.094	.028	.043
Bootstrap $T=1.0$	.123	.087	.035	.038	.035	.136	.106	.020	.024	.023
Bootstrap $T=1.5$	.162	.152	.047	.047	.047	.128	.121	.015	.016	.014
Constrained label P (very high)	.081	.045	.085	.091	.087	.094	.045	.024	.021	.023
Direct (numeric)	.512	.176	.110	.109	.110	.566	.230	.009	.009	.009
MCMC accept $T=0.125$	.464	.113	.103	.103	.103	.519	.169	.009	.009	.009
MCMC agreement $T=0.5$	.338	.208	.050	.054	.055	.378	.189	.034	.009	.013
Steering sensitivity	.425	.138	.089	.090	.090	.471	.189	.003	.003	.003

Table 9: Post-hoc calibration on Gemma-3-27B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

M Training the Qwen3.6-27B activation oracle

The Qwen3-8B oracle is the released checkpoint of Karvonen et al. (2025). For Qwen3.6-27B no oracle existed prior to this work. We adapt their LatentQA trainer to a hybrid Gated DeltaNet (3/4 of token-mixing sublayers) plus Gated Attention (1/4 of sublayers) architecture and train a verbalizer with the same data mixture (system-prompt question answering, binary classification, self-supervised context prediction) and the same LoRA recipe ($r=8$, one epoch, bf16). Two changes were needed:

LoRA target selection. The upstream regex matches only `q_proj|k_proj|v_proj|o_proj` and the MLP projections. Those names exist only in the 1/4 full-attention sublayers of Qwen3.6. The 3/4 Gated DeltaNet sublayers expose Linear modules under different names (`in_proj_qkv`, `in_proj_z`, `in_proj_b`, `in_proj_a`, `out_proj`). Without expanding the regex, LoRA wraps only one quarter of the token-mixing capacity and training loss flattens well above the homogeneous baseline. Our expanded regex covers both attention

types while still excluding the vision tower and `lm_head`.

Activation-collection slicing. The upstream multi-layer activation collector had an off-by-one in its offset window. We patched the slicing to honor `[max_offset:min_offset]` consistently, which matters when collecting from the trailing tokens of long inputs.

After these two fixes the verbalizer trains to comparable out-of-distribution validation loss as the homogeneous Qwen3-8B oracle (one B200, ≈ 36 hours wall-clock, single epoch on $\approx 1M$ examples). The matching secret-word target adapters are trained with the recipe of section L.

N Hardware, software, and runtime

All runs use a single NVIDIA B200 GPU with bfloat16 weights. The Qwen3-8B benchmark sweep completes in ≈ 8 hours wall-clock; the Qwen3.6-27B sweep in ≈ 48 hours; the two Gemma sweeps and the forced-choice, constrained-label, and extended-temperature backfills add ≈ 70 hours combined. Software: PyTorch 2.11, Hugging Face Transformers 5.8, PEFT 0.19, flash-attn

k	Qwen3-8B ($T=1.0$)			Qwen3.6-27B ($T=1.5$)			Gemma-2-9B ($T=1.5$)			Gemma-3-27B ($T=1.3$)		
	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC
3	.285	.240	.808	.107	.313	.735	.112	.307	.698	.298	.223	.814
5	.334	.131	.823	.138	.204	.766	.141	.198	.745	.357	.110	.824
10	.371	.069	.835	.173	.131	.797	.184	.101	.739	.404	.097	.853
20	.402	.057	.829	.206	.103	.812	.214	.054	.748	.422	.117	.876

Table 10: Bootstrap sample-count ablation: mode frequency over the first k of the 20 saved draws, at the ECE-optimal temperature of the original grid for each oracle. Most of the calibration and ranking gain is in place by $k=10$ (on Gemma-3-27B the ECE minimum sits at $k=10$); k is the main cost lever of the bootstrap.

Word	Qwen3-8B	Qwen3.6-27B	Gemma-2-9B	Gemma-3-27B
moon	.827	.380	.623	.467
snow	.727	.413	.320	.487
jump	.627	.240	.633	.553
smile	.610	.240	.623	.423
ship	.580	.100	.263	.473
chair	.557	.190	.133	.513
flag	.523	.097	.513	.477
salt	.477	.277	.060	.527
green	.473	.450	.213	.497
dance	.350	.197	.043	.470
book	.337	.267	.153	.323
gold	.327	.247	.047	.300
cloud	.307	.163	.073	.593
flame	.297	.147	.070	.170
leaf	.240	.050	.087	.373
blue	.180	.500	.157	.560
wave	.170	.110	.000	.467
clock	.170	.157	.277	.277
song	.150	.100	.000	.047
rock	.120	.170	.087	.587

Table 11: Per-word accuracy on Bootstrap $T=1.0$ for all four oracles, sorted by Qwen3-8B accuracy. Per-word accuracy spans an order of magnitude on every oracle.

2.8.3, Python 3.13.

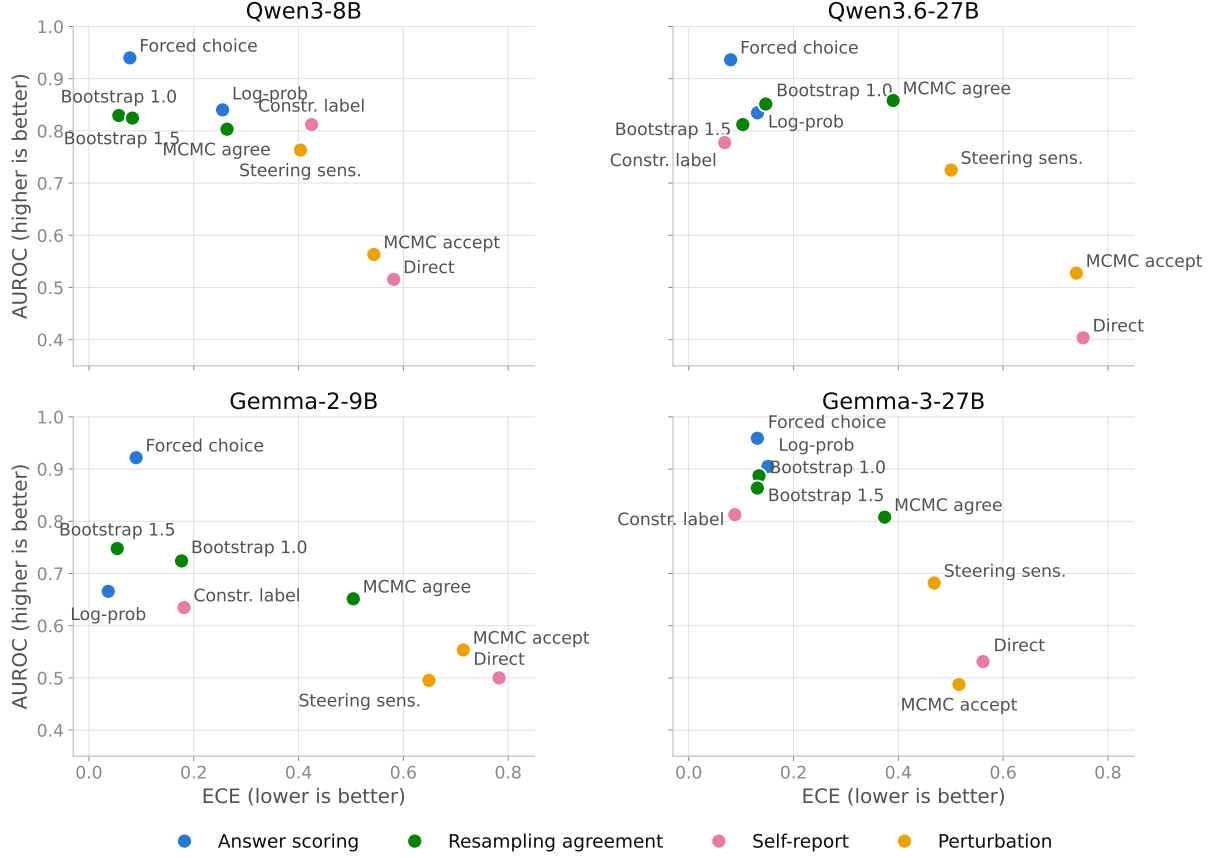


Figure 2: Calibration versus ranking on all four oracles. Each point is one confidence method; the best corner is top left. Colour groups methods by mechanism; every point is labeled directly. Forced choice and the bootstrap family occupy the good corner on every oracle; direct numeric self-report and raw MCMC acceptance sit in the high-ECE, near-chance corner on every oracle.

Method \ N	Qwen3-8B				Qwen3.6-27B			
	2	5	10	20	2	5	10	20
Bootstrap $T=1.0$ ECE	0.213	0.045	0.049	0.057	0.130	0.139	0.155	0.147
Bootstrap $T=0.7$ ECE	0.139	0.108	0.080	0.097	0.194	0.196	0.230	0.218
Bootstrap $T=1.5$ ECE	0.244	0.080	0.084	0.083	0.067	0.084	0.106	0.103
Log-prob (offset) ECE	0.487	0.245	0.275	0.255	0.157	0.160	0.119	0.131
MCMC agree $T=0.5$ ECE	0.047	0.269	0.242	0.263	0.350	0.374	0.395	0.390
Direct (numeric) ECE	0.291	0.584	0.563	0.582	0.710	0.719	0.764	0.753
Bootstrap $T=1.0$ AUROC	0.795	0.791	0.819	0.829	0.842	0.837	0.852	0.851
Bootstrap $T=0.7$ AUROC	0.783	0.801	0.815	0.830	0.854	0.847	0.871	0.863
Log-prob (offset) AUROC	0.829	0.815	0.828	0.840	0.835	0.820	0.847	0.835
MCMC agree $T=0.5$ AUROC	0.775	0.753	0.786	0.803	0.866	0.834	0.865	0.858

Table 12: Controlled target-set scaling on the two Qwen oracles. Bootstrap $T=1.0$ on Qwen3-8B is the only method whose ECE improves as N grows; most others degrade. On Qwen3.6-27B every method is roughly N -stable.

Method	Qwen3-8B			Qwen3.6-27B			Gemma-2-9B			Gemma-3-27B		
	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC
MCMC agreement $T=0.5$	.413	.263 $\pm .005$	.803 $\pm .006$	.228	.390 $\pm .005$	.858 $\pm .006$	.214	.504 $\pm .006$	.652 $\pm .008$	.435	.374 $\pm .005$	.808 $\pm .005$
Steering sensitivity	.418	.404 $\pm .006$	.763 $\pm .005$	.232	.501 $\pm .005$	.725 $\pm .007$	.208	.649 $\pm .006$	.495 $\pm .008$	.433	.469 $\pm .006$	.682 $\pm .005$
MCMC accept $T=0.125$	.415	.544 $\pm .006$	.563 $\pm .005$	.234	.740 $\pm .006$	.528 $\pm .004$	.217	.714 $\pm .005$	.553 $\pm .006$	.436	.516 $\pm .007$	.487 $\pm .007$

Table 13: The three confidence readouts built for the steered setting, $n=6,000$ samples per row per oracle, at the best temperature of each grid. ECE and AUROC carry $\pm$ one bootstrap standard error (1000 resamples); best ECE and AUROC per oracle among these three in **bold**. Every row here is weaker on both axes than the tuned bootstrap of table 1.

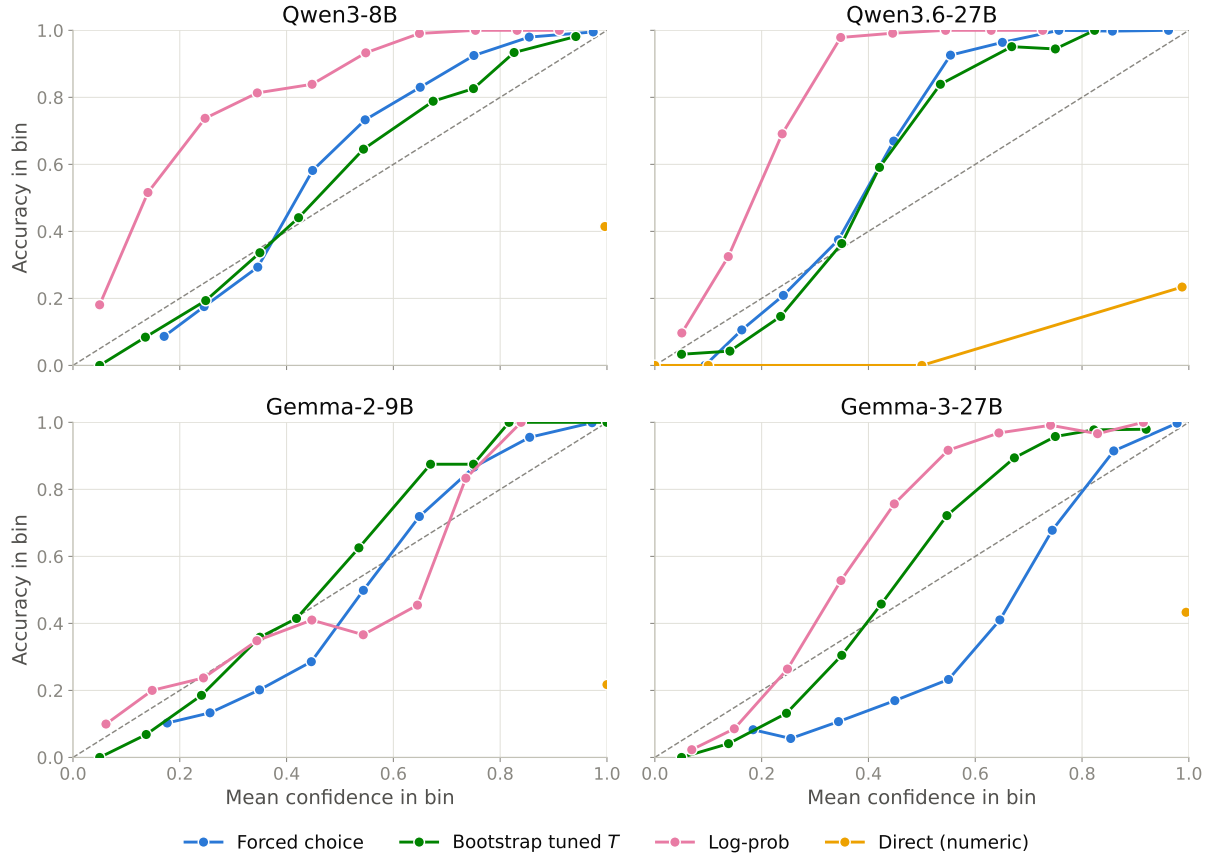


Figure 3: Reliability diagrams on all four oracles. The dashed diagonal is perfect calibration; bins with no samples are omitted. Direct numeric self-report appears as one or two points at confidence near 1 because the oracle reports the same high number on almost every sample.

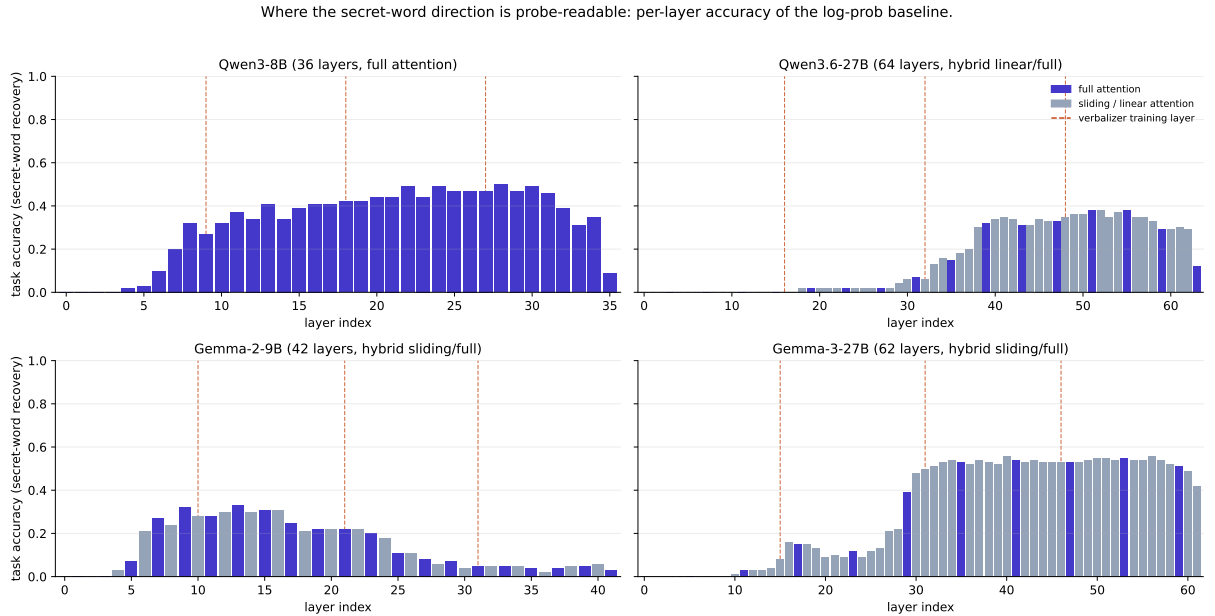


Figure 4: Per-layer log-prob accuracy on the secret-word task across all decoder layers of four oracle bases. Dashed lines mark the three layers each verbalizer was trained on; bar shading distinguishes full-attention (indigo) from sliding/linear-attention (slate) layers. Band peaks: Qwen3-8B 50% at layer 28; Qwen3.6-27B 38% at layer 51; Gemma-2-9B 33% at layer 13; Gemma-3-27B 56% at layer 40 (broad plateau ~ 50 –56% across layers 31–59). Gemma-2-9B reads early; the Qwens read late; Gemma-3-27B has the widest probe-readable band of the four.

T	α	Qwen3-8B			Qwen3.6-27B		
		Acc	ECE	AUR.	Acc	ECE	AUR.
.500	2	.376	.551	.601	.211	.746	.574
.250	4	.403	.547	.579	.230	.738	.547
.125	8	.415	.544	.563	.234	.740	.528

Table 14: M6 (power-sampling acceptance) temperature sweep on the two Qwen oracles; the Gemma rows are in section A. AUR.: AUROC. As T drops the chain accepts more on correct and wrong inputs alike, shrinking the gap between the two pools.

Method	Acc	ECE	Br.	AUR.
<i>Qwen3-8B</i>				
Bootstrap $T=1.0$	.402	.057	.163	.829
MCMC agree $T=0.5$	.413	.263	.247	.803
<i>Qwen3.6-27B</i>				
Bootstrap $T=1.0$	.225	.147	.130	.851
MCMC agree $T=0.5$	.228	.390	.274	.858

Table 15: Bootstrap versus power-sampling agreement at the best temperature of each. *Br.*: Brier. *AUR.*: AUROC.

T	α	Qwen3-8B			Qwen3.6-27B		
		Acc	ECE	AUR.	Acc	ECE	AUR.
.500	2	.413	.263	.803	.228	.390	.858
.250	4	.418	.408	.737	.233	.569	.767
.125	8	.415	.493	.669	.236	.668	.653

Table 16: M7 (power-sampling agreement) temperature sweep. The $\alpha = 2$ row is competitive with bootstrap on AUROC, especially on Qwen3.6-27B (AUR. 0.858), at $5\times$ the wall-clock cost and with weaker ECE.